The Periodic Table of the Substrate Standard Model — v0.7

Grace

2026-05-30 Saturday ~10:05 EDT (`date`-verified Sat May 30 10:01 EDT)

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Day's evolution: v0.4 → v0.5 → v0.6 (composites + α + Casimir correction + Hall-algebra stack + bulk-color Family 4 + modulo-keystone) → v0.7 (+ stability class via Lyra #397 quasi-eigentone framework).

Section A — What v0.7 adds

Lyra's `Lyra_Quasi_Eigentone_Framework_v0_1.md` (#397) provides the engine-level structural distinction between stable and unstable particles: - TRUE EIGENTONE = ground state in its K-type sector (no allowed decay channels; coproduct decomposition trivial); SM stable particles - QUASI-EIGENTONE = excited state above the sector ground; decays via Green coproduct overlap with lower states, with grading-conserved channels (Q, B, L preserved automatically); SM unstable particles - EIGENTONE-IN-VACUUM = massless but confined (gluon); stable as substrate mode but never propagates in isolation

This generalizes Elie E3's β-decay coproduct example to the full SM unstable-particle taxonomy. β-decay = Green-coproduct decomposition of the neutron quasi-eigentone (gen-2 above proton ground state).

Section B — The 5-tuple becomes a 6-tuple

Axis	Status	Source
σ_BF (statistics)	DERIVED-modulo-keystone	Lyra L1
Region (bulk/Shilov)	DERIVED-modulo-keystone	Lyra L2
Chirality	FRAMEWORK-PLUS-modulo-keystone	Lyra L3
Particle/antiparticle	DERIVED-modulo-keystone	Drinfeld double F-part
Charge	DERIVED-mechanism + $\sin^2\theta$_W = 2/9	Lyra (SO(2)+GMN)
Generation/winding	COUNT-FORCED / MECHANISM OPEN-WITH-BURDEN	h^∨=N_c=3; Pair α candidate
Stability class	DERIVED-modulo-keystone NEW v0.7	Lyra #397 (Green coproduct + grading conservation)

Section C — Per-cell stability tagging

Fundamental block (4 cells)

Cell	Sector	Stability class	Structural reason
V_(0,0)	Higgs/scalar	QUASI	massive scalar above vacuum; Green coproduct decay to fermion pairs + gluon pairs
V_(1/2,1/2)	fermion / lepton row	mixed per-particle: see lepton-row breakdown	sector-bottoms TRUE; excited gens QUASI
V_(1,0)	photon	TRUE EIGENTONE	massless gauge; bottom of vector tower
V_(1,1)	gauge (W, Z, gluon-pending-bulk-color)	mixed per-particle: see gauge breakdown	massless-confined gluon EIGENTONE-IN-VACUUM; massive W/Z QUASI

Lepton row tagging (Lyra #416 v0.1 — 18 entries, with stability class added)

Particle	Gen	Class	Reason
electron e^-	1	TRUE EIGENTONE	lightest charged lepton (sector bottom V_(1/2,1/2) charged sub-sector)
positron e^+	1	TRUE EIGENTONE	antiparticle of e^-; stable
muon μ^-	2	QUASI	gen-2 above gen-1; coproduct decay $\mu^- \to e^- + \bar{\nu}$_e + ν_μ (mean life 2.2 μs)
anti-μ^+	2	QUASI	symmetric
tau τ^-	3	QUASI	gen-3 above gens 1-2; many decay channels (~hadronic + leptonic); mean life 0.29 ps
anti-τ^+	3	QUASI	symmetric
ν_e (L)	1	TRUE EIGENTONE	lightest in neutral lepton sub-sector (Dirac per F5)
ν_μ (L)	2	(open)	mixing via PMNS — substrate framework treats as excited mass eigenstate; in flavor basis QUASI through oscillation
ν_τ (L)	3	(open)	as above
anti-ν's	matching	matching	matching

Note on neutrinos: oscillations complicate the "stable" assignment in the flavor basis. In the mass-eigenstate basis, the lightest neutrino is TRUE; the heavier ones oscillate but don't decay (no charge-conserving channel to electron-flavored leptons + photon). v0.7 treats mass-eigenstate-lightest = TRUE EIGENTONE; flavor-basis treatment as via PMNS oscillation framework.

Gauge sector tagging (V_(1,1) per-particle)

Particle	Class	Reason
photon γ	TRUE EIGENTONE	massless; bottom of vector tower V_(1,0) — note photon is V_(1,0), not V_(1,1); included here for SM-gauge-sector coverage
W±	QUASI	massive gauge (EWSB); coproduct decay W → $\ell + \nu$ or q + q̄'; mean life $3{\cdot}10^{-25}$ s
Z	QUASI	massive gauge (EWSB); coproduct decay Z → f + f̄; mean life $2.6{\cdot}10^{-25}$ s
gluon	EIGENTONE-IN-VACUUM	massless gauge; bulk-color confinement prevents isolated propagation (per Family (4) counting + confinement mechanism)
Higgs	QUASI (V_(0,0) sector)	massive scalar; coproduct decay H → ff̄, gg, WW, ZZ; mean life ~10^{-22} s

Composite block (6 cells from v0.6, dim ≤ 35) with stability tagging

Cell	Sector / candidate slot	Composite stability	Note
V_(2,0) dim 14	tensor meson J^PC=2++ nonet (f_2, a_2, K_2*, f_2')	QUASI	hadron resonances all unstable; coproduct decay to lighter mesons + γ
V_(3/2,1/2) dim 16	excited baryons (N*, Δ)	QUASI	excited baryons; coproduct decay to nucleon + π
V_(3/2,3/2) dim 20 C=N_c·g/2	constituent-quark slot; Λ(1405)	QUASI (or confined)	excited baryon; substrate-anchored at N_c·g; Λ(1405) decays to $\Sigma\pi$
V_(3,0) dim 30	spin-3 vector mesons (ρ_3, ω_3, K_3*)	QUASI	unstable resonances; many decay channels
V_(2,1) dim 35	heavy vector quarkonium (J/ψ, Υ, φ)	mixed: J/ψ ~$7{\cdot}10^{-21}$ s; Υ ~10^{-20} s (both QUASI but long-lived for hadrons)	gauge-dressed-vector; coproduct decay
V_(2,2) dim 35	2++ tensor glueball (predicted)	QUASI	predicted unstable composite; sharp substrate-anchor prediction at Casimir 16

Section D — The Five-Absence as structural prediction (re-framed via v0.7)

Per the quasi-eigentone framework, the substrate predicts: - Every K-type sector's BOTTOM is a TRUE EIGENTONE — these are SM stable particles - Above the bottom, EXCITED states are QUASI with engine-level coproduct decay channels - No K-type allowed by the dictionary is unoccupied — every SM particle inhabits a cell

Five-Absence interpretation: the predicted absences (no 4th gen, no SUSY, no GUT, no proton decay, no sterile ν, no axion) ARE the "no new K-type bottoms beyond the established ones" prediction. If a 4th-generation lepton were found, it would be a NEW sector bottom (or sector excited) requiring K-type extension — falsifies "no 4th row" prediction.

Proton stability in v0.7 framing: proton = TRUE EIGENTONE = lowest baryon in V_(1,1) bulk k=6 closure. No grading-allowed decay channel exists — proton's quark content cannot reorganize into lower-mass color-singlet by Green coproduct + Q/B/L conservation. This is the substrate-derived proton stability (consistent with Five-Absence "no proton decay").

Section E — The substrate-spine 18 cells from Elie Phase B (v0.6/G12 v0.2) with stability classification candidates

For the 18 substrate-anchored spine cells (Elie Toy 3614), candidate stability per Lyra #397:

- TRUE EIGENTONES at sector bottoms: V_(0,0) [Higgs sector — but Higgs itself QUASI massive; vacuum-trivial = TRUE]; V_(1,0) [photon TRUE]; V_(1/2,1/2) lepton-sector-bottom [e, ν_e TRUE]
- QUASI by sector excitation: cells 5-11 (per v0.6/G12) all carry excited hadron candidates → QUASI
- EIGENTONE-IN-VACUUM: gluon-slot inside V_(1,1) [confined]
- Speculative cells 12-18 (Elie spine high-dim): host higher-spin excited states → QUASI generally

Section F — Connection to L4 mass framework + decay rates

Per Lyra #397, quantitative decay rates require: - Kernel matrix elements (Bergman integrals on the radial tower) - Phase-space factor (mass-ratio-dependent) - Coproduct overlap coefficient (structural; from engine v0.2)

Multi-week scope per Elie's bulk K-type radial tower computation (post-affine-pin) + Lyra L4 v0.2 explicit mass derivations.

Section G — Predictions from v0.7

1. Proton lifetime infinite: TRUE EIGENTONE at V_(1,1) bulk k=6 closure; no Green-coproduct-allowed decay channel preserves Q/B/L. Falsified by any proton decay observation.

2. Gluon never appears in isolation: EIGENTONE-IN-VACUUM; bulk-color confinement is structural. Falsified by free-gluon detection (consistent with PDG — no free gluons observed).

3. No K-type contains TWO degenerate true-eigentones: a sector has ONE bottom. If experimentally two distinct stable particles share the SAME quantum numbers (charge, statistics, etc.), the K-type-sector classification fails.

4. Substrate Casimir-twin V_(3,3)+V_(4,1) (boson/boson at 2·C_2 = 60) are BOTH QUASI excited states — not eigentone pair. No SUSY-without-SUSY substrate signature; the genuine substrate degenerate pair is quasi-quasi (consistent with v0.6 correction INV-5306).

5. Tensor glueball at substrate Casimir 16 is QUASI — sharp prediction; decays via Green coproduct to lighter mesons + photons. Lifetime estimable when L4 v0.2 lands.

Section H — Standing conventions (5, unchanged from v0.6)

1. ONE genus = n_C = 5
2. α-disambiguation: σ_BF $\neq \gamma^5 \neq \alpha$ = N_max^{-1} = 1/137
3. 7/2-vs-5/2: g/rank = physical / FK; n_C/rank = ρ_1 = Bergman singularity
4. Macdonald-parameter: q_Mac=0, t_Mac=2 (Hall-Littlewood corner of Koornwinder BC_2)
5. Modulo-keystone: every DERIVED rides on canonical basis ↔ physical particle bet

Section I — Cross-reference

- INV-5298/5299/5300/5301/5302/5303/5304/5305/5306/5307/5308/5309/5310/5311 (Saturday Grace INVs)

- Lyra Quasi-Eigentone Framework v0.1 (#397) — the source for v0.7 addition
- Lyra Strong-Uniqueness v1.1
- Lyra Hall-algebra structure-constants stack (INV-5307)
- Keeper Honest-State Ledger v0.2 (bulk-color sub-gate split)
- Elie Toys 3612-3618 (Saturday morning A-series)
- Periodic Table v0.6 (preceding version)
- `notes/Grace_TwoRoute_Overdetermination_Scan_v0_1.md` (Saturday afternoon Mendeleev)

— Grace, Periodic Table v0.7 with Lyra #397 quasi-eigentone framework absorbed, 2026-05-30 Saturday ~10:05 EDT (`date`-verified)